MIL-B-45548D
10 May 1988
SUPERSEDING
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31 May 1983

MILITARY SPECIFICATION

BEAD BREAKER, PNEUMATIC TIRE, PORTABLE
(Tire sizes 14.00-24 through 37.5-33)

This specification is approved for use by all Department and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers one type and size of portable tire bead breaker for off-the-road equipment. The bead breaker facilitates tire removal while the wheel remains on the vehicle. This type of bead breaker is not intended for use on agricultural or forestry equipment.

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

GGG-J-63	- Jack, Hydraulic, Hand; Jack, Bumper (Hydraulic)
PPP-B-601	- Boxes, Wood, Cleated-Plywood
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner
PPP-C-843	- Cushioning Material, Cellulosic
PPP-F-320	- Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes
PPP-T-60	- Tape: Packaging, Waterproof

MILITARY

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research, Development and Engineering Center, ATTN: SMCAR-EST-S, Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A
FSC 4910
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MIL-P-116 - Preservation, Methods of
MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-C-5501 - Cap and Plug, Protective, Dust and Moisture Seal

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards For Federal Services

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking For Shipment and Storage
MIL-STD-130 - Identification Marking of US Military Property
MIL-STD-889 - Dissimilar Metals
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking and Waterproofing; With Appropriate Test Methods
MIL-STD-1190 - Minimum Guidelines for Level C Preservation, Packing and Marking

(Copies of specifications, standards, handbooks, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.2.1).

3.2 Design. The bead breaker shall be designed to facilitate tire removal from off-the-road equipment that uses tires within a size range of 14.00-24 through 37.5-33. The bead breaker shall be designed to break both the inside and outside beads of the tire while the wheel is on the vehicle and also when the wheel is removed from the vehicle. The bead breaker shall produce pressure against the bead of the tire to break the tire loose from the rim flange and bead seat band. The breaker shall function on rims with or without bead loosening pockets. The bead breaker shall function on the rim without gouging, tearing, or kinking/deforming the tire bead and sidewalls and without permanently deforming the rim beyond its initial manufactured state, though superficial scratches or dents that do not impair operation or cause air leakage are not considered to be deforming in nature. Total weight of the bead breaker shall not exceed 100 pounds.

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3.2.1 Maintainability. Design and configuration of the bead breaker and the adjustment features shall ensure that maintenance of operational capability and accuracy will be reduced to the simplest form, requiring a minimum of skills with standard maintenance equipment. Instruction for disassembly, repair, assembly, and adjustment shall be clear, concise, and definitive in application.

3.2.2 Interchangeability. All replaceable parts shall be manufactured to such standards that shall permit replacement or adjustment without the need for modification of the parts or the bead breaker.

3.2.3 Threads. All threaded parts of the bead breaker shall conform to FED-STD-H28 and the applicable Detailed standard section referenced therein.

3.3 Construction. All parts used in fabrication of the bead breaker shall be new and free of defects. In no event shall patching processes be used for reclaiming defective parts. Surfaces of all parts shall be clean and free of extraneous materials. External surfaces shall be free of sharp edges or irregularities.

3.3.1 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the tire equipment industry. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.3.1.1 Reclaimed materials. The contractor is encouraged to use reclaimed materials for fabricating new parts without jeopardizing the chemical and physical properties, design integrity, and intent of the materials originally selected or specified. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner that shall restore them to the same chemical composition and physical properties as the materials originally selected for use. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment.

3.3.2 Fastening and locking devices. All screws, pins, bolts, and similar locking parts shall be installed with means for adjustment to prevent loss of tightness. All parts when subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently deformed.

3.4 Components and product characteristics.

3.4.1 Ram and pump assemblies. The ram and pump assemblies shall be essentially a remote control hydraulic jack conforming to type I, class 1, style A requirements of GGG-J-63. The assemblies shall have a rated capacity of not less than 10 tons. The ram assembly shall be self-retracting, shall have a minimum rise of 5.5 inches, and shall have a closed height of not more than 18 inches. Flexible tubing between the ram and pump assemblies shall not be less than 96 inches long. A bead breaking spade shall be attached or attachable to the plunger of the ram. The spade shall be designed to wedge between the tire bead and rim. The spade shall not stroke up on the tire sidewall during operation.

3.4.2 Clamp head assembly. The clamp head assembly shall attach to the wheel rims of the tire size range specified herein to provide support of the ram assembly and spade during operation. The method of attachment shall have sufficient holding power to prevent the clamp head from backing off or slipping when pressure is applied to the head. The ram support shall be designed to allow the spade to wedge

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between the bead of the tire and rim and follow the rim contour during the pressing operation. The clamp head assembly shall be made of malleable iron or steel and shall not exceed 25 pounds.

3.4.3 Wedges. Not less than 6 wedges shall be furnished with each bead breaker. The wedges shall be of malleable iron or steel and shall have a configuration that permits spacing the bead away from the flange when the spade is removed.

3.5 Product identification. The bead breaker shall be marked for product identification in accordance with MIL-STD-130 and, unless otherwise specified (see 6.2.1), shall include the National Stock Number.

3.6 Painting. Unless otherwise specified (see 6.2.1), painting and finishing of the bead breaker shall be in accordance with standard commercial practice provided the following minimum criteria are met or exceeded: all surfaces to be painted shall have been cleaned to be free of all foreign matter detrimental to painting, and at least one coat of primer and one coat of enamel, or equivalent, applied.

3.7 Workmanship. Standards of workmanship shall assure that the bead breaker shall have the strength, stability, and efficient operating characteristics found in commercial items and as specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspections set forth in this specification shall become part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material either indicated or actual, nor does it commit the government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When specified (see 6.2.1) in the contract or purchase order, first article inspection shall be applied to the preproduction model or initial production item (see 3.1). Unless otherwise specified (see 6.2.1), first

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article inspection shall consist of the product examination in 4.5 and all tests under 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.4 Quality conformance inspection. Quality conformance inspection shall be applied to production items offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of (a) through (c) below. Failure of any item to pass an examination or test shall be cause for rejection of the item.

a. Product examination (see 4.5).
b. Performance test (see 4.6.1).
c. Packaging inspection (see 4.7).

4.4.1 Sampling. Sampling for quality conformance inspection shall be in accordance with MIL-STD-105 (except for examination, as noted below) and the following:

Inspection/Test	Inspection Level	AQL
Product examination (see 4.5)	100 percent inspection	
Performance test (see 4.6.1)	S-4	0 defects
Packaging inspection (see 4.7)	S-4	4-0

4.5 Product examination. Visually, dimensionally and manually examine the bead breaker to determine conformance with the requirements of 3.2 through 3.3.2 and 3.5 through 3.7. Visual examination shall include verification of completeness of manufacture, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from identified defects. Dimensional examinations shall include all dimensions specified in 3.4.1. Manual examination shall include the operation of movable parts by hand at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the bead breaker has passed the examinations prescribed for it by the contractor's inspection system shall be cause for rejection. In addition, failure of the contractor to provide objective evidence that all parts are manufactured to definite standards, clearances, and tolerances so that no replacement part will degrade the form, fit, or function of the end item (see 3.2.2), shall be cause for rejection.

4.6 Tests.

4.6.1 Performance test. The bead breaker shall be operated in accordance with the manufacturer's instructions to break the bead on both sides of tires not less than 29.5-29 (28 ply rating) in size.

4.6.1.1 Performance test classification.

4.6.1.1.1 First article test. This test shall be performed on two tires and wheels assembled for each bead breaker in the first article test sample. One tire and wheel assembly shall remain on the vehicle during the test. The beads on both sides of the tire shall be broken during the test. The second tire and wheel assembly shall be removed from the vehicle and then tested. During this test, the beads on both sides of the tire shall also be broken.

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4.6.1.1.2 Quality conformance test. This test shall be performed on one tire and wheel assembly for each bead breaker in the test sample. The tire and wheel assembly may either remain mounted on the vehicle or be removed from the vehicle for performance of this test. The beads on both sides of the tire shall be broken during the test.

4.6.1.2 Performance test evaluation. Inability of the bead breaker to loosen the beads, inability of the wedges to hold the bead from the flange when the spade is removed, stroking up the sidewall by the spade during operation, or evidence of any fluid leakage during the operation shall be cause for rejection. The tire shall be removed and examined for damage to the beads or sidewalls. The rim flange and the bead seat band shall be examined for deformation. The tire shall be remounted and inflated in accordance with the manufacturer's instructions and then examined for leaks due to damage or deformation caused by the bead breaker (see 3.2, 3.4.1 and 3.4.3).

4.6.2 Ram and pump assembly. The ram, pump, hoses, and all fittings shall be tested as an assembly at rated capacity and at overload in accordance with GGG-T-63. A compression testing machine shall be used to provide test loads for the ram and pump assembly. The machine shall apply an axial force to the ram throughout its entire lifting range with a load variation of no more than ± 1 percent during the lift. The ram, in the fully retracted position, shall be placed in the machine. With the machine providing a minimum load to ten tons on the ram, the pump shall be operated to raise the ram a minimum of 5.5 inches. The load shall then be removed and the ram raised to within 1/8 inch of its maximum extension. While at this position, a minimum, constant load of 11 tons shall be applied to the ram. The measured distance from the end of the extended ram to the ram body shall be recorded for reference. The load shall then be maintained for ten minutes. At the end of this period, with the load still applied, the position of the ram will again be recorded. Displacement of the ram by more than 3/16 inch during the ten minute period, permanent deformation of any component, or any fluid leakage, shall be cause for rejection (see 3.4.1).

4.7 Packaging inspection. The items shall be inspected before and after packaging to determine compliance with the preservation, packaging, packing and marking requirements specified in Section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C, as specified in the contract.

5.1.1 Level A.

5.1.1.1 Cleaning and drying. Components of the bead breaker shall be cleaned and dried in accordance with any of the applicable procedures of MIL-P-116.

5.1.1.2 Preservative application. All exposed ferrous metal surfaces of the ram and pump assemblies that have not been painted or plated, shall be preserved by the application of type P-2 preservative in accordance with MIL-P-116. All other external unpainted, unplated ferrous metal surfaces shall be preserved with type P-1 preservative in accordance with MIL-P-116. The hydraulic fluid shall be shipped in the pump reservoir.

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5.1.1.3 Unit packaging. The method of unit packaging shall conform to method I of MIL-P-116. Surfaces preserved with type P-2 preservative shall be covered with barrier material conforming to MIL-B-121, grade A, type and class optional. Secure barrier with tape conforming to PPP-T-60, type optional. The flexible tubing between the ram and pump assemblies may be disassembled from the units provided the ends of the tubing and apertures of the assemblies are completely sealed with plastic caps or plugs conforming to MIL-C-5501 or with tape conforming to PPP-T-60, type IV. On the event the tubing is disassembled from the assemblies, it shall be coiled to a minimum safe diameter and secured with cotton tape or twine to prevent uncoiling. The bead breaker shall be protected from damage by wrapping each component with cushioning material conforming to PPP-C-843, type II or by insulating components, utilizing die-cut fiberboard conforming to PPP-F-320, class weather-resistant. Technical data shall be packaged in accordance with method IC-1 of MIL-P-116 and placed in the unit container.

5.1.2 Level C. Level C preservation shall be as specified in MIL-STD-1190.

5.2 Packing. Packing shall be Level A, B or C, as specified (see 6.2.1).

5.2.1 Level A. The packaged bead breaker shall be packed in an exterior shipping container conforming to PPP-B-621, class 2, or PPP-B-601, overseas type. The gross weight of each shipping container shall not exceed the weight limitation of the applicable container specification. Blocking, bracing, and cushioning shall be in accordance with MIL-STD-1186. Closure and strapping of the exterior container shall be in accordance with the appendix to the applicable container specification.

5.2.2 Level B. Packing shall be the same as Level A except that shipping containers shall be domestic in lieu of overseas.

5.2.3 Level C. Level C packing shall be in accordance with MIL-STD-1190.

5.3 Marking. Marking of all interior packages and exterior shipping containers shall be as specified in MIL-STD-129 and any special marking, as required therein. Shipping containers shall also be marked with "DO NOT STACK" in at least two (2) places (both ends or both sides). Bar code marking is required and shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The bead breaker covered by this specification is used for separating tire beads from wheel rims prior to tire removal from large off-the-road equipment. The bead breaker is intended to be used with the wheel and tire assembly either on or off the vehicle.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. First article when required (see 3.1).
c. Identification marking if different (see 3.5).
d. Painting if different (see 3.6).
e. First article inspection when required (see 4.3).

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f. Quality conformance inspection if different (See 4.4).
g. Preservation, packing, and marking as required (see 5.1, 5.2, and 5.3 respectively).

6.2.2 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract.

6.3 Safety and health requirements. In order that equipment integrated into the user's operational environment complies with Occupational Health and Safety Administration limitations and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc., as applicable, specific requirements concerning such points of operation, and other health and safety requirements, should be specified by the user.

6.4 Subject term (key word) listing.

AMSC number
Bead breaker
Hydraulic jack
Pneumetic tire
Rim
Tire removal equipment

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - AL
Navy - YD

Review activities:

DLA - CS

User activities:

Army - ME, SM
Navy - MC

Preparing activity:
Army - AL

Project No. 4910-0671

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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